

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

Сеть	Устройство	IP-адрес
192.168.10.0/24	R4	192.168.10.1
	PC1	192.168.10.10
192.168.11.0/24	R4	192.168.11.1
	R5	192.168.11.2
192.168.12.0/24	R5	192.168.12.1
	R1	192.168.12.2
192.168.13.0/24	R5	192.168.13.1
	PC2	192.168.13.10

Сеть	Устройство	IP-адрес
192.168.0.0/24	R1	192.168.0.1
	R2	192.168.0.2
	R3	192.168.0.3

Сеть	Устройство	IP-адрес
192.168.30.0/24	R6	192.168.30.1
	PC3	192.168.30.10
192.168.31.0/24	R7	192.168.31.1
	PC4	192.168.31.10

192.168.32.0/24	R6	192.168.32.1
	R7	192.168.32.2
192.168.33.0/24	R2	192.168.33.1
	R6	192.168.33.2
192.168.34.0/24	R3	192.168.34.1
	R7	192.168.34.2

Сеть	Устройство	IP-адрес
192.168.20.0/24	R8	192.168.20.1
	PC5	192.168.20.10
192.168.21.0/24	R3	192.168.21.1
	R8	192.168.21.2

Настройка адресов:

```
PC1> ip 192.168.10.10 24 192.168.10.1
```

Checking for duplicate address...

```
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1
```

```
R4#enable
```

```
R4#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R4(config)#int fa0/0
```

```
R4(config-if)#ip address 192.168.10.1 255.255.255.0
```

```
R4(config-if)#no sh
```

```
R4(config-if)#exit
```

```
R4(config)#
```

*Mar 1 00:02:42.599: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:02:43.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R4(config)#int fa1/0

R4(config-if)#ip address 192.168.11.1 255.255.255.0

R4(config-if)#no sh

R4(config-if)#exit

R4(config)#exit

R5#enable

R5#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#int fa0/0

R5(config-if)#ip address 192.168.13.1 255.255.255.0

R5(config-if)#no sh

R5(config-if)#

*Mar 1 00:04:13.159: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:04:14.159: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R5(config-if)#exit

R5(config)#int fa1/0

R5(config-if)#ip address 192.168.11.2 255.255.255.0

R5(config-if)#no sh

R5(config-if)#

*Mar 1 00:05:28.519: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:05:29.519: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R5(config-if)#exit

R5(config)#int fa2/0

R5(config-if)#ip address 192.168.12.1 255.255.255.0

R5(config-if)#no sh

R5(config-if)#exit

R5(config)#

*Mar 1 00:06:01.019: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up

*Mar 1 00:06:02.019: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R5(config)#exit

PC2> ip 192.168.13.10 24 192.168.13.1

Checking for duplicate address...

PC2 : 192.168.13.10 255.255.255.0 gateway 192.168.13.1

R1#enable

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#int fa0/0

R1(config-if)#ip address 192.168.12.2 255.255.255.0

R1(config-if)#no sh

R1(config-if)#

*Mar 1 00:05:55.283: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:05:56.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit

R1(config)#int fa1/0

R1(config-if)#ip address 192.168.0.1 255.255.255.0

R1(config-if)#no sh

R1(config-if)#

*Mar 1 00:06:34.927: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:06:35.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R1(config-if)#exit

R1(config)#exit

R2#enable

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#int fa0/0

R2(config-if)#ip address 192.168.0.2 255.255.255.0

R2(config-if)#no sh

R2(config-if)#

*Mar 1 00:06:47.623: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:06:48.623: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up

R2(config-if)#exit

R2(config)#int fa1/0

R2(config-if)#ip address 192.168.33.1 255.255.255.0

R2(config-if)#no sh

R2(config-if)#

*Mar 1 00:07:39.559: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:07:40.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up

R2(config-if)#exit

R2(config)#exit

R3#enable

R3#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#int fa0/0

R3(config-if)#ip address 192.168.0.3 255.255.255.0

R3(config-if)#no sh

R3(config-if)#

*Mar 1 00:07:47.135: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:07:48.135: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R3(config-if)#exit

R3(config)#int fa1/0

R3(config-if)#ip address 192.168.34.1 255.255.255.0

R3(config-if)#no sh

R3(config-if)#

*Mar 1 00:08:39.271: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:08:40.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config-if)#exit

R3(config)#int fa2/0

R3(config-if)#ip address 192.168.21.1 255.255.255.0

R3(config-if)#no sh

R3(config-if)#

*Mar 1 00:09:27.727: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up

*Mar 1 00:09:28.727: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R3(config-if)#exit

R3(config)#exit

R8#enable

R8#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R8(config)#int fa0/0

R8(config-if)#ip address 192.168.21.2 255.255.255.0

R8(config-if)#no sh

R8(config-if)#

*Mar 1 00:09:07.151: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:09:08.151: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R8(config-if)#exit

R8(config)#int fa1/0

R8(config-if)#ip address 192.168.20.1 255.255.255.0

R8(config-if)#no sh

R8(config-if)#exit

*Mar 1 00:09:45.451: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:09:46.451: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R8(config-if)#exit

R8(config)#exit

PC5> ip 192.168.20.10 24 192.168.20.1

Checking for duplicate address...

PC5 : 192.168.20.10 255.255.255.0 gateway 192.168.20.1

R6#enable

R6#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#int fa0/0

R6(config-if)#ip address 192.168.33.2 255.255.255.0

R6(config-if)#no sh

R6(config-if)#

*Mar 1 00:10:25.359: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:10:26.359: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R6(config-if)#exit

R6(config)#int fa1/0

R6(config-if)#ip address 192.168.30.1 255.255.255.0

R6(config-if)#no sh

R6(config-if)#e

*Mar 1 00:11:09.055: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:11:10.055: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R6(config-if)#exit

R6(config)#int fa2/0

R6(config-if)#ip address 192.168.32.1 255.255.255.0

R6(config-if)#no sh

R6(config-if)#ex

*Mar 1 00:11:49.167: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up

*Mar 1 00:11:50.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R6(config-if)#exit

R6(config)#exit

R7#enable

R7#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R7(config)#int fa0/0

R7(config-if)#ip address 192.168.34.2 255.255.255.0

R7(config-if)#no sh

R7(config-if)#

*Mar 1 00:11:38.267: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:11:39.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R7(config-if)#exit

R7(config)#int fa1/0

R7(config-if)#ip address 192.168.31.1 255.255.255.0


```
R7(config-if)#no sh
```

```
R7(config-if)#
```

```
*Mar 1 00:12:17.923: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
```

```
*Mar 1 00:12:18.923: %LINEPROTO-5-UPDOWN: Line protocol on Interface  
FastEthernet1/0, changed state to up
```

```
R7(config-if)#exit
```

```
R7(config)#int fa2/0
```

```
R7(config-if)#ip address 192.168.32.2 255.255.255.0
```

```
R7(config-if)#no sh
```

```
R7(config-if)#
```

```
*Mar 1 00:12:45.363: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
```

```
*Mar 1 00:12:46.363: %LINEPROTO-5-UPDOWN: Line protocol on Interface  
FastEthernet2/0, changed state to up
```

```
R7(config-if)#exit
```

```
R7(config)#exit
```

```
PC3> ip 192.168.30.10 24 192.168.30.1
```

```
Checking for duplicate address...
```

```
PC3 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1
```

```
PC4> ip 192.168.31.10 24 192.168.31.1
```

```
Checking for duplicate address...
```

```
PC4 : 192.168.31.10 255.255.255.0 gateway 192.168.31.1
```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

```
R1#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#router rip
```

```
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 192.168.12.0
R1(config-router)#exit
R1(config)#exit
```

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.10.0
R4(config-router)#network 192.168.11.0
R4(config-router)#exit
R4(config)#exit
```

```
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#network 192.168.11.0
R5(config-router)#network 192.168.12.0
R5(config-router)#network 192.168.13.0
R5(config-router)#exit
R5(config)#exit
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#router ospf 1

R1(config-router)#network 192.168.0.0 0.0.0.255 area 0

R1(config-router)#exit

R1(config)#exit

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router ospf 1

R2(config-router)#network 192.168.0.0 0.0.0.255 area 0

R2(config-router)#

*Mar 1 00:32:27.059: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.12.2 on FastEthernet0/0 from LOADING to FULL, Loading Done

R2(config-router)#network 192.168.33.0 0.0.0.255 area 2

R2(config-router)#exit

R2(config)#exit

R3#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router ospf 1

R3(config-router)#network 192.168.0.0 0.0.0.255 area 0

R3(config-router)#network 192.168.34.0 0.0.0.255 area 2

R3(config-router)#

*Mar 1 00:33:25.311: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.33.1 on FastEthernet0/0 from LOADING to FULL, Loading Done

R3(config-router)#network 192.168.21.0 0.0.0.255 area 1

*Mar 1 00:33:26.819: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.12.2 on FastEthernet0/0 from LOADING to FULL, Loading Done

```
R3(config-router)#network 192.168.21.0 0.0.0.255 area 1
```

```
R3(config-router)#area 1 stub no-summary
```

```
R3(config-router)#exit
```

```
R3(config)#exit
```

```
R6#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R6(config)#router ospf 1
```

```
R6(config-router)#network 192.168.33.0 0.0.0.255 area 2
```

```
R6(config-router)#network 192.168.32.0 0.0.0.255 area 2
```

```
R6(config-router)#
```

```
*Mar 1 00:31:14.747: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.33.1 on FastEthernet0/0  
from LOADING to FULL, Loading Done
```

```
R6(config-router)#network 192.168.30.0 0.0.0.255 area 2
```

```
R6(config-router)#exit
```

```
R6(config)#exit
```

```
R7#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R7(config)#router ospf 1
```

```
R7(config-router)#network 192.168.34.0 0.0.0.255 area 2
```

```
R7(config-router)#network 192.168.32.0 0.0.0.255 area 2
```

```
R7(config-router)#
```

```
*Mar 1 00:30:33.947: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.34.1 on FastEthernet0/0  
from LOADING to FULL, Loading Done
```

```
R7(config-router)#
```

```
*Mar 1 00:30:36.123: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.33.2 on FastEthernet2/0  
from LOADING to FULL, Loading Done
```

```
R7(config-router)#network 192.168.31.0 0.0.0.255 area 2
```

```
R7(config-router)#exit
```

```
R7(config)#exit
```

```
R8#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R8(config)#router ospf 1
```

```
R8(config-router)#network 192.168.21.0 0.0.0.255 area 1
```

```
R8(config-router)#network 192.168.20.0 0.0.0.255 area 1
```

```
R8(config-router)#area 1 stub
```

```
R8(config-router)#exit
```

```
R8(config)#exit
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

```
R1#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
R1(config)#router ospf 1
```

```
R1(config-router)#redistribute rip subnets
```

```
R1(config-router)#
```

```
*Mar 1 00:01:01.423: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.33.1 on FastEthernet1/0  
from LOADING to FULL, Loading Done
```

```
R1(config-router)#exit
```

```
R1(config)#router rip
```

```
R1(config-router)#redistribute ospf 1 metric 5
```

```
R1(config-router)#exit
```

```
R1(config)#exit
```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

От PC1:

PC1> ping 192.168.13.10

192.168.13.10 icmp_seq=1 timeout

84 bytes from 192.168.13.10 icmp_seq=2 ttl=62 time=21.591 ms

84 bytes from 192.168.13.10 icmp_seq=3 ttl=62 time=25.610 ms

84 bytes from 192.168.13.10 icmp_seq=4 ttl=62 time=25.362 ms

84 bytes from 192.168.13.10 icmp_seq=5 ttl=62 time=26.055 ms

PC1> ping 192.168.30.10

192.168.30.10 icmp_seq=1 timeout

84 bytes from 192.168.30.10 icmp_seq=2 ttl=59 time=59.279 ms

84 bytes from 192.168.30.10 icmp_seq=3 ttl=59 time=56.355 ms

84 bytes from 192.168.30.10 icmp_seq=4 ttl=59 time=55.424 ms

84 bytes from 192.168.30.10 icmp_seq=5 ttl=59 time=65.609 ms

PC1> ping 192.168.31.10

192.168.31.10 icmp_seq=1 timeout

84 bytes from 192.168.31.10 icmp_seq=2 ttl=59 time=58.747 ms

84 bytes from 192.168.31.10 icmp_seq=3 ttl=59 time=66.014 ms

84 bytes from 192.168.31.10 icmp_seq=4 ttl=59 time=55.437 ms

84 bytes from 192.168.31.10 icmp_seq=5 ttl=59 time=55.660 ms

PC1> ping 192.168.20.10

192.168.20.10 icmp_seq=1 timeout

84 bytes from 192.168.20.10 icmp_seq=2 ttl=59 time=58.295 ms

84 bytes from 192.168.20.10 icmp_seq=3 ttl=59 time=56.282 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=59 time=55.483 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=59 time=55.953 ms

От PC2:

PC2> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=62 time=22.288 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=62 time=25.030 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=62 time=25.666 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=62 time=25.487 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=62 time=25.812 ms

PC2> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=60 time=50.312 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=60 time=45.910 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=60 time=45.753 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=60 time=45.717 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=60 time=45.379 ms

PC2> ping 192.168.31.10

84 bytes from 192.168.31.10 icmp_seq=1 ttl=60 time=42.859 ms
84 bytes from 192.168.31.10 icmp_seq=2 ttl=60 time=45.512 ms
84 bytes from 192.168.31.10 icmp_seq=3 ttl=60 time=46.023 ms
84 bytes from 192.168.31.10 icmp_seq=4 ttl=60 time=45.624 ms
84 bytes from 192.168.31.10 icmp_seq=5 ttl=60 time=45.353 ms

PC2> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=60 time=41.889 ms

84 bytes from 192.168.20.10 icmp_seq=2 ttl=60 time=45.883 ms

84 bytes from 192.168.20.10 icmp_seq=3 ttl=60 time=45.471 ms

84 bytes from 192.168.20.10 icmp_seq=4 ttl=60 time=46.277 ms

84 bytes from 192.168.20.10 icmp_seq=5 ttl=60 time=55.976 ms

От PC3:

PC3> ping 192.168.13.10

84 bytes from 192.168.13.10 icmp_seq=1 ttl=60 time=50.147 ms

84 bytes from 192.168.13.10 icmp_seq=2 ttl=60 time=45.872 ms

84 bytes from 192.168.13.10 icmp_seq=3 ttl=60 time=45.563 ms

84 bytes from 192.168.13.10 icmp_seq=4 ttl=60 time=46.504 ms

84 bytes from 192.168.13.10 icmp_seq=5 ttl=60 time=46.231 ms

PC3> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=81.336 ms

84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=75.992 ms

84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=66.116 ms

84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=65.761 ms

84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=55.980 ms

PC3> ping 192.168.31.10

84 bytes from 192.168.31.10 icmp_seq=1 ttl=62 time=33.273 ms
84 bytes from 192.168.31.10 icmp_seq=2 ttl=62 time=25.579 ms
84 bytes from 192.168.31.10 icmp_seq=3 ttl=62 time=25.402 ms
84 bytes from 192.168.31.10 icmp_seq=4 ttl=62 time=26.069 ms
84 bytes from 192.168.31.10 icmp_seq=5 ttl=62 time=25.539 ms

PC3> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=60 time=61.627 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=60 time=45.211 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=60 time=46.108 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=60 time=56.040 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=60 time=56.199 ms

От PC4:

PC4> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=68.881 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=75.276 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=65.471 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=66.158 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=55.636 ms

PC4> ping 192.168.13.10

84 bytes from 192.168.13.10 icmp_seq=1 ttl=60 time=81.764 ms
84 bytes from 192.168.13.10 icmp_seq=2 ttl=60 time=55.462 ms
84 bytes from 192.168.13.10 icmp_seq=3 ttl=60 time=45.642 ms

84 bytes from 192.168.13.10 icmp_seq=4 ttl=60 time=55.396 ms

84 bytes from 192.168.13.10 icmp_seq=5 ttl=60 time=55.771 ms

PC4> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=38.585 ms

84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=25.578 ms

84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=25.259 ms

84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=25.808 ms

84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=25.532 ms

PC4> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=61 time=49.267 ms

84 bytes from 192.168.20.10 icmp_seq=2 ttl=61 time=35.861 ms

84 bytes from 192.168.20.10 icmp_seq=3 ttl=61 time=35.990 ms

84 bytes from 192.168.20.10 icmp_seq=4 ttl=61 time=35.057 ms

84 bytes from 192.168.20.10 icmp_seq=5 ttl=61 time=35.078 ms

От PC5:

PC5> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=66.789 ms

84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=55.184 ms

84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=65.323 ms

84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=55.897 ms

84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=56.188 ms

PC5> ping 192.168.13.10

84 bytes from 192.168.13.10 icmp_seq=1 ttl=60 time=59.547 ms

84 bytes from 192.168.13.10 icmp_seq=2 ttl=60 time=45.254 ms

84 bytes from 192.168.13.10 icmp_seq=3 ttl=60 time=45.900 ms

84 bytes from 192.168.13.10 icmp_seq=4 ttl=60 time=45.518 ms

84 bytes from 192.168.13.10 icmp_seq=5 ttl=60 time=46.012 ms

PC5> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=60 time=53.841 ms

84 bytes from 192.168.30.10 icmp_seq=2 ttl=60 time=46.123 ms

84 bytes from 192.168.30.10 icmp_seq=3 ttl=60 time=45.673 ms

84 bytes from 192.168.30.10 icmp_seq=4 ttl=60 time=45.342 ms

84 bytes from 192.168.30.10 icmp_seq=5 ttl=60 time=45.972 ms

PC5> ping 192.168.31.10

84 bytes from 192.168.31.10 icmp_seq=1 ttl=61 time=48.667 ms

84 bytes from 192.168.31.10 icmp_seq=2 ttl=61 time=36.013 ms

84 bytes from 192.168.31.10 icmp_seq=3 ttl=61 time=36.180 ms

84 bytes from 192.168.31.10 icmp_seq=4 ttl=61 time=35.192 ms

84 bytes from 192.168.31.10 icmp_seq=5 ttl=61 time=35.621 ms

6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

R4-R5:

rip						
No.	Time	Source	Destination	Protocol	Length	Info
3	15.962923	192.168.11.1	224.0.0.9	RIPv2	66	Response
4	16.057253	192.168.11.2	224.0.0.9	RIPv2	106	Response
7	26.550572	192.168.11.2	224.0.0.9	RIPv2	186	Response
21	84.966405	192.168.11.2	224.0.0.9	RIPv2	246	Response
22	87.085983	192.168.11.1	224.0.0.9	RIPv2	66	Response
50	153.907457	192.168.11.2	224.0.0.9	RIPv2	246	Response
51	155.271268	192.168.11.1	224.0.0.9	RIPv2	66	Response
76	219.961216	192.168.11.2	224.0.0.9	RIPv2	246	Response
77	222.958773	192.168.11.1	224.0.0.9	RIPv2	66	Response
86	290.153597	192.168.11.2	224.0.0.9	RIPv2	246	Response
87	293.968850	192.168.11.1	224.0.0.9	RIPv2	66	Response
102	351.890470	192.168.11.2	224.0.0.9	RIPv2	246	Response
105	361.156434	192.168.11.1	224.0.0.9	RIPv2	66	Response
122	419.260819	192.168.11.2	224.0.0.9	RIPv2	246	Response
123	422.800299	192.168.11.1	224.0.0.9	RIPv2	66	Response
130	486.516875	192.168.11.1	224.0.0.9	RIPv2	66	Response
131	486.738774	192.168.11.2	224.0.0.9	RIPv2	246	Response
149	550.656621	192.168.11.2	224.0.0.9	RIPv2	246	Response
150	551.391204	192.168.11.1	224.0.0.9	RIPv2	66	Response
156	614.233388	192.168.11.2	224.0.0.9	RIPv2	246	Response
158	620.459447	192.168.11.1	224.0.0.9	RIPv2	66	Response
165	678.023351	192.168.11.2	224.0.0.9	RIPv2	246	Response
167	684.196470	192.168.11.1	224.0.0.9	RIPv2	66	Response

R4 посылает Response-сообщения на мультикаст о своей сети с PC1:

>	Frame 51: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface -, id 0
>	Ethernet II, Src: cc:04:68:e9:00:10 (cc:04:68:e9:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
>	Internet Protocol Version 4, Src: 192.168.11.1, Dst: 224.0.0.9
>	User Datagram Protocol, Src Port: 520, Dst Port: 520
▼	Routing Information Protocol
	Command: Response (2)
	Version: RIPv2 (2)
▼	IP Address: 192.168.10.0, Metric: 1
	Address Family: IP (2)
	Route Tag: 0
	IP Address: 192.168.10.0
	Netmask: 255.255.255.0
	Next Hop: 0.0.0.0
	Metric: 1

R5 посылает Response-сообщения о своей сети с R1 и всех сетях дальше:

```
> Frame 76: Packet, 246 bytes on wire (1968 bits), 246 bytes captured (1968 bits) on interface -, id 0
> Ethernet II, Src: cc:05:69:07:00:10 (cc:05:69:07:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
> Internet Protocol Version 4, Src: 192.168.11.2, Dst: 224.0.0.9
> User Datagram Protocol, Src Port: 520, Dst Port: 520
  Routing Information Protocol
    Command: Response (2)
    Version: RIPv2 (2)
    > IP Address: 192.168.0.0, Metric: 6
    > IP Address: 192.168.12.0, Metric: 1
    > IP Address: 192.168.13.0, Metric: 1
    > IP Address: 192.168.20.0, Metric: 6
    > IP Address: 192.168.21.0, Metric: 6
    > IP Address: 192.168.30.0, Metric: 6
    > IP Address: 192.168.31.0, Metric: 6
    > IP Address: 192.168.32.0, Metric: 6
    > IP Address: 192.168.33.0, Metric: 6
    > IP Address: 192.168.34.0, Metric: 6
```

Информацией о сети между R4 и R5 они не обмениваются.

R1-Switch:

ospf						
No.	Time	Source	Destination	Protocol	Length	Info
3	0.814809	192.168.0.1	224.0.0.5	OSPF	90	Hello Packet
12	10.766265	192.168.0.2	224.0.0.5	OSPF	90	Hello Packet
16	10.814538	192.168.0.1	224.0.0.5	OSPF	94	Hello Packet
17	10.816363	192.168.0.2	192.168.0.1	OSPF	78	DB Description
18	10.816373	192.168.0.2	192.168.0.1	OSPF	94	Hello Packet
20	10.824624	192.168.0.1	192.168.0.2	OSPF	78	DB Description
21	10.824647	192.168.0.1	192.168.0.2	OSPF	178	DB Description
23	10.826732	192.168.0.2	192.168.0.1	OSPF	198	DB Description
24	10.834724	192.168.0.1	192.168.0.2	OSPF	78	DB Description
25	10.836635	192.168.0.2	192.168.0.1	OSPF	78	DB Description
26	10.836644	192.168.0.2	192.168.0.1	OSPF	118	LS Request
27	10.844810	192.168.0.1	192.168.0.2	OSPF	130	LS Request
28	10.844825	192.168.0.1	192.168.0.2	OSPF	78	DB Description
29	10.844828	192.168.0.1	192.168.0.2	OSPF	242	LS Update
30	10.846811	192.168.0.2	192.168.0.1	OSPF	238	LS Update
31	10.846821	192.168.0.2	224.0.0.5	OSPF	98	LS Update
32	10.854890	192.168.0.1	192.168.0.2	OSPF	78	LS Acknowledge
33	10.887001	192.168.0.2	224.0.0.5	OSPF	206	LS Update
34	10.895081	192.168.0.1	192.168.0.2	OSPF	138	LS Acknowledge
35	11.040554	192.168.0.3	224.0.0.5	OSPF	98	Hello Packet
36	11.045983	192.168.0.1	192.168.0.3	OSPF	78	DB Description
37	11.046010	192.168.0.1	192.168.0.3	OSPF	98	Hello Packet
38	11.050624	192.168.0.3	192.168.0.1	OSPF	78	DB Description
39	11.056084	192.168.0.1	192.168.0.3	OSPF	298	DB Description
40	11.060649	192.168.0.3	192.168.0.1	OSPF	318	DB Description
41	11.066170	192.168.0.1	192.168.0.3	OSPF	78	DB Description
42	11.070770	192.168.0.3	192.168.0.1	OSPF	78	DB Description
43	11.070782	192.168.0.3	192.168.0.1	OSPF	142	LS Request
44	11.076252	192.168.0.1	192.168.0.3	OSPF	154	LS Request
45	11.076274	192.168.0.1	192.168.0.3	OSPF	78	DB Description
46	11.076278	192.168.0.1	192.168.0.3	OSPF	274	LS Update
47	11.080888	192.168.0.3	192.168.0.1	OSPF	294	LS Update
48	11.319538	192.168.0.2	224.0.0.5	OSPF	98	LS Update
49	11.322297	192.168.0.3	224.0.0.5	OSPF	98	LS Update
50	11.358100	192.168.0.1	224.0.0.5	OSPF	98	LS Update
51	11.573875	192.168.0.3	224.0.0.5	OSPF	98	LS Update
52	11.614079	192.168.0.3	224.0.0.5	OSPF	98	LS Update
56	13.380361	192.168.0.1	224.0.0.5	OSPF	398	LS Acknowledge
57	13.596067	192.168.0.3	224.0.0.5	OSPF	218	LS Acknowledge
58	13.603518	192.168.0.2	224.0.0.5	OSPF	278	LS Acknowledge
61	16.298056	192.168.0.1	192.168.0.2	OSPF	98	LS Update
62	16.300473	192.168.0.2	192.168.0.1	OSPF	78	LS Acknowledge
66	20.787834	192.168.0.2	224.0.0.5	OSPF	98	Hello Packet
67	20.795612	192.168.0.1	224.0.0.5	OSPF	98	Hello Packet
68	21.042231	192.168.0.3	224.0.0.5	OSPF	98	Hello Packet
76	30.778901	192.168.0.2	224.0.0.5	OSPF	98	Hello Packet
77	30.818646	192.168.0.1	224.0.0.5	OSPF	98	Hello Packet
78	31.034186	192.168.0.3	224.0.0.5	OSPF	98	Hello Packet
89	40.779774	192.168.0.2	224.0.0.5	OSPF	98	Hello Packet
90	40.818998	192.168.0.1	224.0.0.5	OSPF	98	Hello Packet

Все маршрутизаторы отправляют Hello-пакеты на мультикаст, после чего начинают обмениваться о сетях которые знают через DB Description, LS Request, LS Update и LS Acknowledge.

DB Description от R1 к R2:

```
> Frame 21: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface -, id 0
> Ethernet II, Src: cc:01:68:8f:00:10 (cc:01:68:8f:00:10), Dst: cc:02:68:ad:00:00 (cc:02:68:ad:00:00)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.2
▼ Open Shortest Path First
  ▼ OSPF Header
    Version: 2
    Message Type: DB Description (2)
    Packet Length: 132
    Source OSPF Router: 192.168.12.2
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0xaf56 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▼ OSPF DB Description
    Interface MTU: 1500
    > Options: 0x52, (O) Opaque, (L) LLS Data block, (E) External Routing
    > DB Description: 0x02, (M) More
    DD Sequence: 2409
  ▼ LSA-type 1 (Router-LSA), len 36
    .000 0000 0011 1011 = LS Age (seconds): 59
    0... .... .... = Do Not Age Flag: 0
    > Options: 0x22, (DC) Demand Circuits, (E) External Routing
    LS Type: Router-LSA (1)
    Link State ID: 192.168.12.2
    Advertising Router: 192.168.12.2
    Sequence Number: 0x80000001
    Checksum: 0xd403
    Length: 36
  ▼ LSA-type 5 (AS-External-LSA (ASBR)), len 36
    .000 0000 0011 0111 = LS Age (seconds): 55
    0... .... .... = Do Not Age Flag: 0
    > Options: 0x20, (DC) Demand Circuits
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.10.0
    Advertising Router: 192.168.12.2
    Sequence Number: 0x80000001
    Checksum: 0x5b5a
    Length: 36
  ▼ LSA-type 5 (AS-External-LSA (ASBR)), len 36
    .000 0000 0011 0111 = LS Age (seconds): 55
    0... .... .... = Do Not Age Flag: 0
    > Options: 0x20, (DC) Demand Circuits
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.11.0
    Advertising Router: 192.168.12.2
    Sequence Number: 0x80000001
    Checksum: 0x5064
    Length: 36
  ▼ LSA-type 5 (AS-External-LSA (ASBR)), len 36
    .000 0000 0011 0011 = LS Age (seconds): 51
    0... .... .... = Do Not Age Flag: 0
    > Options: 0x20, (DC) Demand Circuits
```

LS Request от R2 к R1:

```

> Frame 26: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: cc:02:68:ad:00:00 (cc:02:68:ad:00:00), Dst: cc:01:68:8f:00:10 (cc:01:68:8f:00:10)
> Internet Protocol Version 4, Src: 192.168.0.2, Dst: 192.168.0.1
▼ Open Shortest Path First
  ▼ OSPF Header
    Version: 2
    Message Type: LS Request (3)
    Packet Length: 84
    Source OSPF Router: 192.168.33.1
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0x1f46 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▼ Link State Request
    LS Type: Router-LSA (1)
    Link State ID: 192.168.12.2
    Advertising Router: 192.168.12.2
  ▼ Link State Request
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.10.0
    Advertising Router: 192.168.12.2
  ▼ Link State Request
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.11.0
    Advertising Router: 192.168.12.2
  ▼ Link State Request
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.12.0
    Advertising Router: 192.168.12.2
  ▼ Link State Request
    LS Type: AS-External-LSA (ASBR) (5)
    Link State ID: 192.168.13.0
    Advertising Router: 192.168.12.2

```

LS Update от R1 к R2:


```

> Frame 29: Packet, 242 bytes on wire (1936 bits), 242 bytes captured (1936 bits) on interface -, id 0
> Ethernet II, Src: cc:01:68:8f:00:10 (cc:01:68:8f:00:10), Dst: cc:02:68:ad:00:00 (cc:02:68:ad:00:00)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.2
▼ Open Shortest Path First
  ▼ OSPF Header
    Version: 2
    Message Type: LS Update (4)
    Packet Length: 208
    Source OSPF Router: 192.168.12.2
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0x4f44 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▼ LS Update Packet
    Number of LSAs: 5
    ▼ LSA-type 1 (Router-LSA), len 36
      .000 0000 0011 1100 = LS Age (seconds): 60
      0... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      Link State ID: 192.168.12.2
      Advertising Router: 192.168.12.2
      Sequence Number: 0x80000001
      Checksum: 0xd403
      Length: 36
      > Flags: 0x02, (E) AS boundary router
      Number of Links: 1
      > Type: Stub ID: 192.168.0.0 Data: 255.255.255.0 Metric: 1
    ▼ LSA-type 5 (AS-External-LSA (ASBR)), len 36
      .000 0000 0011 1000 = LS Age (seconds): 56
      0... .... = Do Not Age Flag: 0
      > Options: 0x20, (DC) Demand Circuits
      LS Type: AS-External-LSA (ASBR) (5)
      Link State ID: 192.168.10.0
      Advertising Router: 192.168.12.2
      Sequence Number: 0x80000001
      Checksum: 0x5b5a
      Length: 36
      Netmask: 255.255.255.0
      1... .... = External Type: Type 2 (metric is larger than any other link state path)
      .000 0000 = TOS: 0
      Metric: 20
      Forwarding Address: 0.0.0.0
      External Route Tag: 0
    ▼ LSA-type 5 (AS-External-LSA (ASBR)), len 36
      .000 0000 0011 1000 = LS Age (seconds): 56
      0... .... = Do Not Age Flag: 0
      > Options: 0x20, (DC) Demand Circuits
      LS Type: AS-External-LSA (ASBR) (5)
      Link State ID: 192.168.11.0
      Advertising Router: 192.168.12.2
      -

```

LS Acknowledge от R2 на мульткаст:

```

> Frame 58: Packet, 278 bytes on wire (2224 bits), 278 bytes captured (2224 bits) on interface -, id 0
> Ethernet II, Src: cc:02:68:ad:00:00 (cc:02:68:ad:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 192.168.0.2, Dst: 224.0.0.5
▼ Open Shortest Path First
  ▼ OSPF Header
    Version: 2
    Message Type: LS Acknowledge (5)
    Packet Length: 244
    Source OSPF Router: 192.168.33.1
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0xbfa2 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▼ LSA-type 1 (Router-LSA), len 36
    .000 0000 0011 1100 = LS Age (seconds): 60
    0... .... .... .... = Do Not Age Flag: 0
    > Options: 0x22, (DC) Demand Circuits, (E) External Routing
    LS Type: Router-LSA (1)
    Link State ID: 192.168.34.1
    Advertising Router: 192.168.34.1
    Sequence Number: 0x80000001
    Checksum: 0x545a
    Length: 36
  ▼ LSA-type 3 (Summary-LSA (IP network)), len 28
    .000 0000 0000 0101 = LS Age (seconds): 5
    0... .... .... .... = Do Not Age Flag: 0
    > Options: 0x22, (DC) Demand Circuits, (E) External Routing
    LS Type: Summary-LSA (IP network) (3)
    Link State ID: 192.168.20.0
    Advertising Router: 192.168.34.1
    Sequence Number: 0x80000001
    Checksum: 0x0e23
    Length: 28
  ▼ LSA-type 3 (Summary-LSA (IP network)), len 28
    .000 0000 0011 1000 = LS Age (seconds): 56
    0... .... .... .... = Do Not Age Flag: 0
    > Options: 0x22, (DC) Demand Circuits, (E) External Routing
    LS Type: Summary-LSA (IP network) (3)
    Link State ID: 192.168.21.0
    Advertising Router: 192.168.34.1
    Sequence Number: 0x80000001
    Checksum: 0xf838
    Length: 28
  ▼ LSA-type 3 (Summary-LSA (IP network)), len 28
    .000 0000 0000 0101 = LS Age (seconds): 5
    0... .... .... .... = Do Not Age Flag: 0
    > Options: 0x22, (DC) Demand Circuits, (E) External Routing
    LS Type: Summary-LSA (IP network) (3)
    Link State ID: 192.168.30.0
    Advertising Router: 192.168.34.1
    Sequence Number: 0x80000001

```

7) Сохранить в отдельные файлы с префиксом rt_ и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.